THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET D

Chapters 5 & 6: **Circles, Measuring Space: Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. ABCD is a cyclic quadrilateral with $\angle A = 75^\circ$. Then $\angle C$ is: 1

(a) 53° **(b)** 285° **(c)** 105° **(d)** 75°

2. ABCD is a cyclic quadrilateral with $\angle A = 123^\circ$. Then $\angle C$ is: 1

(a) 29° **(b)** 237° **(c)** 57° **(d)** 123°

3. A perpendicular drawn from the centre of a circle to a chord: 1

(a) is parallel to the chord **(b)** bisects the chord

(c) is equal to the chord **(d)** is equal to the radius

4. Through two given points the number of circles that can be drawn is: 1

(a) exactly two **(b)** infinitely many **(c)** exactly one **(d)** none

5. The perpendicular bisector of a chord of a circle always passes through: 1

(a) no fixed point **(b)** a point outside the circle

(c) the midpoint of the arc only **(d)** the centre of the circle

6. The area of a parallelogram with base 20 cm and height 9 cm is: 1

(a) 90 cm^2 **(b)** 58 cm^2 **(c)** 200 cm^2 **(d)** 180 cm^2

7. By Brahmagupta's formula, the area of a cyclic quadrilateral with sides $3, 4, 3, 4$ is: 1

(a) 12 sq. units **(b)** 16 sq. units **(c)** 24 sq. units **(d)** 10 sq. units

8. Taking $\pi = \frac{22}{7}$, the area of a circle of radius 14 cm is: 1

(a) 1232 cm^2 **(b)** 88 cm^2 **(c)** 616 cm^2 **(d)** 308 cm^2

9. A median of a triangle divides it into two triangles of: 1

(a) one double the other **(b)** equal area

(c) equal angles **(d)** equal perimeter

10. Āryabhaṭa called his value 3.1416 for π 'āsanna', meaning: 1

(a) approximate (approaching) **(b)** exact

(c) wrong **(d)** largest possible

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In quadrilateral ABCD, $\angle A = 100^\circ$ and $\angle C = 80^\circ$. Is it cyclic? Give a reason. 2

12. Side BC of cyclic quadrilateral ABCD is produced to E and $\angle DCE = 70^\circ$. Find $\angle A$. 2

13. Why does an inscribed regular hexagon show that $\pi > 3$? 2

14. Find the perimeter of a quadrant (quarter-circle) of radius 14 cm, including the two radii. ($\pi = \frac{22}{7}$) 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. An arc AB subtends $\angle AOB = (6x - 10)^\circ$ at the centre and $\angle ACB = (2x + 5)^\circ$ at a point C on the circle. Find x. 3

16. Show that Heron's formula is a special case of Brahmagupta's formula. 3

17. The angles A and C of a cyclic quadrilateral are in the ratio 4 : 5. Find them. 3

OR

Explain why three collinear points cannot lie on a circle. 3

18. The minute hand of a clock is 14 cm long. Find the distance its tip travels in 30 minutes and the area swept in that time ($\pi = \frac{22}{7}$). 3

OR

A rectangle has perimeter 36 cm. Find the dimensions of the rectangle with the greatest area and that area. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A cyclic quadrilateral has sides 5, 5, 12, 12 (in order). 5

(i) Show that its diagonal is a diameter and find the radius.

(ii) Find its area.

(iii) Explain why this quadrilateral is not a rectangle.

OR

(i) Prove that an angle inscribed in a semicircle is 90°. 5

(ii) Chord AB subtends $\angle ACB = 62^\circ$ at C on the major arc. Find the angle AB subtends at the centre and at a point D on the minor arc.

20. Baudhāyana's problem: 5

(i) Explain what it means to ‘square a rectangle’.

(ii) Find the side of the square equal in area to a $25\ \text{cm} \times 4\ \text{cm}$ rectangle.

(iii) Find its perimeter and compare it with the rectangle's perimeter.

OR

Case study: A circular park of radius 35 m has a circular fountain of radius 7 m at its centre. ($\pi = \frac{22}{7}$) 5

(i) Find the area of the park.

(ii) Find the area of the fountain.

(iii) Find the area of grass.

(iv) Find the length of railing needed to fence the fountain and the park boundary together.

END OF QUESTION PAPER — **Total: 40 Marks**